

Assignment 5, Part 2: Reinforcement Learning Theory

Setup for both questions. The vocabulary is I, like, pizza. Each episode is 3 words. Only "I like pizza" gives a reward of +10. All other sentences give 0. Gamma = 1.

Question 1

1. Possible 1-word and 2-word states

There are 3 one-word states:

[I], [like], [pizza]

There are $3 \times 3 = 9$ two-word states:

[I, I], [I, like], [I, pizza],
[like, I], [like, like], [like, pizza],
[pizza, I], [pizza, like], [pizza, pizza]

2. Terminal states

Each of the 3 positions has 3 choices, so there are

$$3^3 = 27$$

terminal states. Only 1 of them gives a reward. The other 26 give 0.

3. Value functions

Each word is picked with probability $1/3$. The reward comes only at the end, and gamma = 1. So the value of a state is the chance of ending at "I like pizza" times 10.

$$V(s) = P(\text{reach "I like pizza" from } s) \times 10$$

(a) $s_0 = []$

All 3 words must be right.

$$P = (1/3)(1/3)(1/3) = 1/27$$

$$V(s_0) = 10/27 = 0.370$$

(b) $s_1 = [I]$

The first word is right. Two words are left.

$$P = (1/3)(1/3) = 1/9$$

$$V(s_1) = 10/9 = 1.111$$

(c) $s_2 = [I, \text{like}]$

Only pizza is left.

$$P = 1/3$$

$$V(s_2) = 10/3 = 3.333$$

(d) $s_3 = [I, \text{pizza}]$

The second word is wrong, so the goal can no longer be reached.

$$P = 0$$

$$V(s_3) = 0$$

Summary:

$$V([]) = 10/27 = 0.370$$

$$V([I]) = 10/9 = 1.111$$

$$V([I, \text{like}]) = 10/3 = 3.333$$

$$V([I, \text{pizza}]) = 0$$

The value rises as the agent gets closer to the goal.

Question 2

1. Q-learning update rule

$$Q(s, a) = Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

The term in brackets is the temporal difference error.

2. Updated value of $Q([I], \text{like})$

Given: $Q([I], \text{like}) = 1.0$, $\alpha = 0.5$, $\gamma = 1$.

Taking "like" from $[I]$ leads to $s' = [I, \text{like}]$. That state has 2 words, so the sentence is not finished. The reward is 0.

$$\max_{a'} Q([I, \text{like}], a') = \max(1.0, 0.5, 2.0) = 2.0$$

Substituting:

$$\begin{aligned} Q([I], \text{like}) &= 1.0 + 0.5 [0 + 2.0 - 1.0] \\ &= 1.0 + 0.5 (1.0) \\ &= 1.5 \end{aligned}$$

Answer: $Q([I], \text{like}) = 1.5$